

Smart Attendance System

DevOps Assignment -2

Branch: Information Technology,3rd Year Section 1

Details:-

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Smart Attendance System

Introduction:-

In modern educational institutions, managing student attendance manually is inefficient, time-consuming, and prone to errors such as incorrect entries and proxy attendance. To overcome these limitations, the Smart Attendance System is developed as a web-based solution that automates attendance tracking and reporting. The system ensures accuracy, reduces manual workload, and provides real-time access to attendance data for both faculty and students. It also enhances transparency and helps institutions monitor student performance effectively.

About My Application:

The Smart Attendance System is a Flask-based web application designed to simplify the process of recording, managing, and analyzing student attendance. Faculty members can securely log in and mark attendance for students, while students can view their attendance reports at any time. The application automatically calculates attendance percentages, generates monthly Excel reports with conditional formatting, and sends email alerts to students with low attendance. The system provides an efficient, user-friendly interface that improves productivity and ensures accurate record-keeping.

This project is a responsive web application designed to showcase and manage "Smart Attendance System". The application is built as a static front-end project and is fully containerized using **Docker** to ensure environment consistency. It utilizes a **CI/CD pipeline** via **Jenkins** for automated building and deployment.

Technical Stack

Frontend Technologies

- **HTML5** – Used to structure all web pages such as login, dashboard, attendance marking, and student report interfaces, ensuring clean and semantic layout.
- **CSS3** – Applied for styling the application with responsive design, modern layouts, color themes, and visual enhancements like buttons, tables, and form elements.
- **Bootstrap** – Used to create a responsive and user-friendly interface with pre-built components like navigation bars, forms, cards, and grids for faster development.
- **Jinja2 (Flask Templating Engine)** – Enabled dynamic rendering of HTML pages by integrating backend data into the frontend, allowing real-time display of attendance and reports.

Backend & Other Technologies

Python (Flask Framework) – Serves as the core backend technology handling routing, business logic, session management, and integration between frontend and database.

- **APScheduler** – Used to schedule automated tasks such as sending daily attendance alerts to students with low attendance.
- **smtplib (Email Service)** – Handles sending automated email notifications to students whose attendance falls below the required threshold.
- **OpenPyXL** – Used for generating Excel reports with attendance data, including conditional formatting (color highlighting based on percentage).

Database / Data Storage

- **SQLite** – A lightweight, serverless database used to store student records, faculty details, attendance data, and email logs. It provides efficient data management without requiring external database setup.

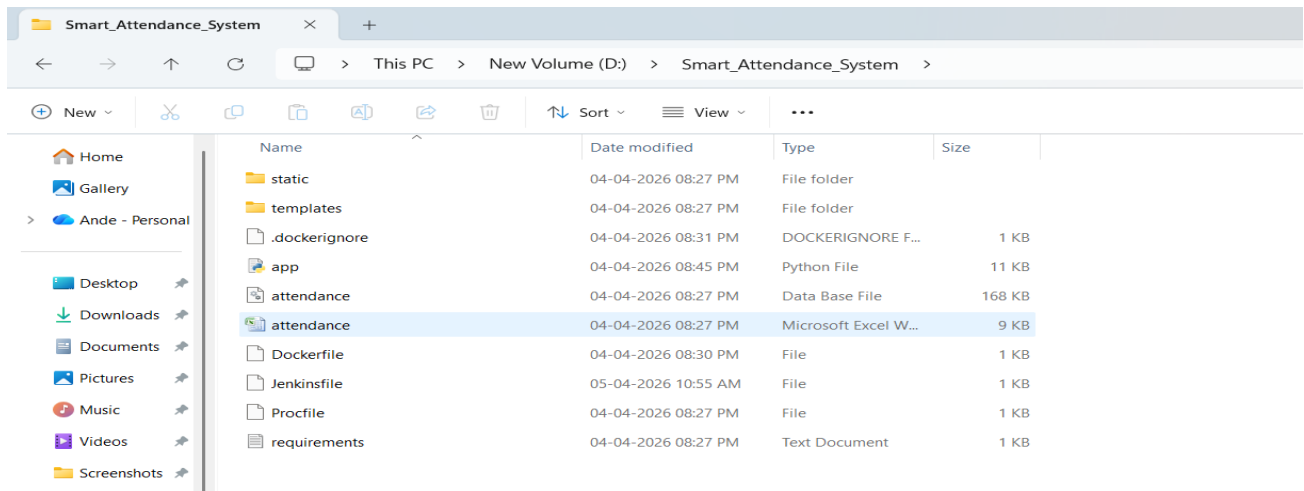
DevOps & Deployment:

- **Docker:** Used to package the application into a lightweight, portable container.
- **Jenkins:** Automation server used to orchestrate the CI/CD pipeline.
- **Nginx (Recommended):** Typically used as the base image in the `Dockerfile` to serve static content.

Architecture of the project:

/Smart-Attendance-System

- |— app.py
- |— requirements.txt
- |— Procfile
- |— attendance.db
- |— attendance.xlsx
- |
- |— /static
 - |— style.css
- |— /templates
 - |— base.html
 - |— index.html
 - |— login.html
 - |— faculty_login.html
 - |— college_login.html
 - |— dashboard.html
 - |— mark_attendance.html
 - |— student_report.html
 - |— success.html



Components:

Inside the application, reusable and structured UI components are organized within the **templates** and **static** folders. These components form the building blocks of the user interface and help maintain consistency across the application.

Base.html Code:

```
<!DOCTYPE html>

<html>

<head>

<title>{% block title %} {% endblock %}</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">

</head>

<body>

<nav class="navbar navbar-dark bg-dark px-3">

    <span class="navbar-brand">Attendance System</span>

    <div>

        <a href="/dashboard" class="btn btn-light btn-sm">Dashboard</a>

        <a href="/logout" class="btn btn-danger btn-sm">Logout</a>

    </div>

</nav>

<div class="container mt-4">

{% block content %} {% endblock %}

</div>

</body>

</html>
```

college login.html Code:

```
<!DOCTYPE html>

<html>

<head>

<title>College Login</title>
```

```

<link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>

<body>

<div class="container">

<h2>College Login</h2>

<form method="POST">

<input type="text" name="college_id" placeholder="College ID" required>

<input type="password" name="password" placeholder="Password" required>

<button type="submit">Login</button>

</form>

{% if error %}

<p class="error">{{ error }}</p>

{% endif %}

</div>

</body>

</html>

```

faculty_login.htmlCode:

```

<!DOCTYPE html>
<html>
<head>
<title>Faculty Login</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

<style>
body {
background: linear-gradient(135deg, #667eea, #764ba2);
}

.glass {
background: rgba(255,255,255,0.15);
padding: 30px;
border-radius: 15px;
color: white;
}
</style>
</head>

```

```

<body class="d-flex justify-content-center align-items-center vh-100">

<div class="glass text-center">
<h3>Faculty Login</h3>

<form method="POST">
<input type="text" name="faculty_code" class="form-control my-2" placeholder="Code">
<input type="password" name="password" class="form-control my-2" placeholder="Password">
<button class="btn btn-light w-100">Login</button>
</form>

{% if error %}
<p class="text-danger">{{ error }}</p>
{% endif %}
</div>

</body>
</html>

```

mark_attendance.html Code:

```

<!DOCTYPE html>

<html>

<head>

<title>Mark Attendance</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

<style>
body {
background: linear-gradient(135deg, #43cea2, #185a9d);
color: white;
font-family: 'Poppins', sans-serif;
}
.container-box {
background: rgba(255,255,255,0.12);
backdrop-filter: blur(12px);
border-radius: 20px;

```

```
padding: 30px;  
}
```

```
.grid-container {  
  display: grid;  
  grid-template-columns: repeat(auto-fill, minmax(260px, 1fr));  
  gap: 15px;  
  max-height: 55vh;  
  overflow-y: auto;  
}
```

```
.student-card {  
  background: rgba(255,255,255,0.2);  
  border-radius: 12px;  
  padding: 12px;  
  display: flex;  
  gap: 10px;  
  cursor: pointer;  
  transition: 0.3s;  
}
```

```
.student-card:hover {  
  background: rgba(255,255,255,0.35);  
}
```

```
.student-card.active {  
  background: #ff6b6b !important;  
}
```

```
.student-info {
```

```
display: flex;
    flex-direction: column;
}

.top-bar {
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.stats {
    display: flex;
    gap: 20px;
    margin-bottom: 15px;
}

.stat-box {
    background: rgba(255,255,255,0.2);
    padding: 10px 20px;
    border-radius: 10px;
}

</style>
</head>

<body>

<div class="container-fluid px-4">

<!-- HEADER -->
```

```
<div class="top-bar mt-3">
```

```
  <h2> 📋 Mark Attendance</h2>
```

```
  <div>
```

```
    <a href="{{ url_for('dashboard') }}" class="btn btn-light me-2">⬅ Back</a>
```

```
    <a href="{{ url_for('logout') }}" class="btn btn-danger">Logout</a>
```

```
  </div>
```

```
</div>
```

```
<div class="container-box mt-3">
```

```
  <!-- 🔍 SEARCH -->
```

```
  <input type="text" id="searchBox" class="form-control mb-3"
```

```
    placeholder=" 🔍 Search student...">
```

```
  <!-- 📊 LIVE STATS -->
```

```
  <div class="stats">
```

```
    <div class="stat-box">Present: <b id="presentCount">0</b></div>
```

```
    <div class="stat-box">Absent: <b id="absentCount">0</b></div>
```

```
  </div>
```

```
  <!-- ACTION BUTTONS -->
```

```
  <div class="mb-3">
```

```
    <button type="button" class="btn btn-light btn-sm" onclick="selectAll()">Select  
All</button>
```

```
    <button type="button" class="btn btn-light btn-sm" onclick="clearAll()">Clear All</button>
```

```
  </div>
```

```
<form method="POST" onsubmit="return confirmSubmit()">
```

<label>Select Hour:</label>

<select name="hour" class="form-control mb-3">

<option value="1">Hour 1</option>

<option value="2">Hour 2</option>

<option value="3">Hour 3</option>

<option value="4">Hour 4</option>

<option value="5">Hour 5</option>

<option value="6">Hour 6</option>

</select>

<h5>Select Absent Students</h5>

<div class="grid-container">

{% for student in students %}

<label class="student-card" data-name="{{ student.name|lower }}">

<input type="checkbox" name="absent" value="{{ student.student_id }}">

<div class="student-info">

{{ student.student_id }}

{{ student.name }}

</div>

</label>

{% endfor %}

</div>

<button class="btn btn-light w-100 mt-4">

 Submit Attendance

```
        </button>

    </form>

</div>

</div>

<script>

    const checkboxes = document.querySelectorAll("input[type='checkbox']");
    const presentCount = document.getElementById("presentCount");
    const absentCount = document.getElementById("absentCount");

    function updateStats() {
        let absent = 0;

        checkboxes.forEach(cb => {
            if (cb.checked) absent++;
        });

        absentCount.innerText = absent;
        presentCount.innerText = checkboxes.length - absent;
    }

    checkboxes.forEach(cb => {
        cb.addEventListener("change", function () {
            this.closest(".student-card").classList.toggle("active", this.checked);
            updateStats();
        });
    });

    function selectAll() {
        checkboxes.forEach(cb => {
            cb.checked = true;
            cb.closest(".student-card").classList.add("active");
        });
    }

```

```
updateStats();
}

function clearAll() {
  checkboxes.forEach(cb => {
    cb.checked = false;
    cb.closest(".student-card").classList.remove("active");
  });
  updateStats();
}

document.getElementById("searchBox").addEventListener("keyup", function () {
  let value = this.value.toLowerCase();

  document.querySelectorAll(".student-card").forEach(card => {
    let name = card.getAttribute("data-name");
    card.style.display = name.includes(value) ? "flex" : "none";
  });
});

function confirmSubmit() {
  return confirm("Are you sure you want to submit attendance?");
}

updateStats();
</script>
</body>
</html>
```

student_report.html Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Student Report</title>

<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap"
rel="stylesheet">

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>

body{

margin:0;

font-family:Poppins;

background:linear-gradient(135deg,#0f172a,#1e293b);

color:white;

}

/* HEADER */

.header{

display:flex;

justify-content:space-between;

align-items:center;

padding:20px 40px;

}

.logout-btn{

background:#ef4444;

color:white;

border:none;

padding:10px 20px;

border-radius:10px;
```

```
cursor:pointer;
transition:0.3s;
}
.logout-btn:hover{
transform:scale(1.1);
}
/* CENTER */
.container{
display:flex;
flex-direction:column;
align-items:center;
}
/* SEARCH BOX */
.search-box{
background:rgba(255,255,255,0.1);
padding:20px;
border-radius:20px;
width:400px;
margin-top:20px;
text-align:center;
backdrop-filter:blur(10px);
}
input{
width:100%;
padding:12px;
border-radius:10px;
border:none;
margin-bottom:10px;
}
```

```
button{  
width:100%;  
padding:12px;  
border:none;  
border-radius:10px;  
background:#38bdf8;  
color:white;  
cursor:pointer;  
}
```

```
/* PROFILE CARD */
```

```
.profile{  
background:linear-gradient(135deg,#1f2937,#111827);  
padding:20px;  
border-radius:15px;  
margin-top:20px;  
text-align:center;  
width:400px;  
box-shadow:0 10px 30px rgba(0,0,0,0.5);  
}
```

```
/* STATS */
```

```
.stats{  
background:rgba(255,255,255,0.1);  
padding:20px;  
border-radius:20px;  
margin-top:20px;  
text-align:center;  
width:400px;  
}
```

```
/* STATUS */
```

```
.good{  
background:#22c55e;  
padding:10px;  
border-radius:10px;  
margin-top:10px;  
}
```

```
.bad{  
background:#ef4444;  
padding:10px;  
border-radius:10px;  
margin-top:10px;  
}
```

```
/* GRAPH */
```

```
.chart{  
width:650px;  
margin-top:30px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<!-- HEADER -->
```

```
<div class="header">
```

```
  <h2>📊 Student Report</h2>
```

```
  <a href="{{ url_for('logout') }}">
```

```
    <button class="logout-btn">Logout</button>
```

```
</a>
```

```
</div>

<div class="container">

<!-- SEARCH -->

<div class="search-box">

<form method="POST">

<input name="student_id" placeholder="Enter Student ID">

<button>Get Report</button>

</form>

</div>

{% if error %}

<p style="color:red">{{error}}</p>

{% endif %}

<!-- PROFILE -->

{% if student %}

<div class="profile">

<h2>👤 {{student.name}}</h2>

<p>ID: {{student.student_id}}</p>

</div>

{% endif %}

<!-- STATS -->

{% if total %}

<div class="stats">

<h3>Total Classes: {{total}}</h3>

<h3>Present: {{present}}</h3>

<h3>Percentage: {{percentage}}%</h3>

{% if percentage >= 75 %}

<div class="good">✅ Good Attendance</div>

{% else %}
```

```
<div class="bad"> ⚠ Low Attendance</div>

{% endif %}

</div>

{% endif %}

<!-- GRAPH -->

{% if subjects %}

<div class="chart">

<canvas id="myChart"></canvas>

</div>

{% endif %}

</div>

<script>

{% if subjects %}

const ctx = document.getElementById('myChart');

new Chart(ctx, {
  type: 'bar',
  data: {
    labels: {{subjects | tojson}},
    datasets: [{
      label: 'Hours Present',
      data: {{present_counts | tojson}},
    }]
  }
});

{% endif %}

</script>

</body>

</html>
```

success.html Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Success</title>

  <!-- Bootstrap -->

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

  <!-- Google Font -->

  <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap"
rel="stylesheet">


  <!-- Confetti Library -->

  <script src="https://cdn.jsdelivr.net/npm/canvas-
confetti@1.6.0/dist/confetti.browser.min.js"></script>


<style>

  body {

    font-family: 'Poppins', sans-serif;

    background: linear-gradient(135deg, #00c9ff, #92fe9d);

    height: 100vh;

    display: flex;

    justify-content: center;

    align-items: center;

  }


  .box {

    background: white;

    padding: 40px;

    border-radius: 25px;
```

```
text-align: center;

box-shadow: 0 15px 40px rgba(0,0,0,0.2);

animation: fadeIn 0.8s ease-in-out;
}
```

```
.tick {

font-size: 60px;

color: green;

animation: pop 0.5s ease;
}
```

```
@keyframes pop {

0% {transform: scale(0);}

100% {transform: scale(1);}

}

@keyframes fadeIn {

from {opacity: 0; transform: translateY(20px);}

to {opacity: 1; transform: translateY(0);}

}
```

```
.countdown {

font-size: 20px;

margin-top: 10px;

color: #555;

}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="box">
```

```
  <div class="tick"> ✓ </div>
```

```
  <h2 class="text-success mt-3">Attendance Submitted Successfully!</h2>
```

```
  <p class="mt-2">You will be logged out shortly...</p>
```

```
  <div class="countdown">
```

```
    Redirecting in <span id="timer">2</span> seconds ⌚
```

```
  </div>
```

```
</div>
```

```
<script>
```

```
  // 🎉 Confetti Effect
```

```
  confetti({
    particleCount: 150,
    spread: 90,
    origin: { y: 0.6 }
  });
```

```
  // ⌚ Countdown Timer
```

```
  let timeLeft = 5;
  const timer = document.getElementById("timer");
```

```
  const countdown = setInterval(() => {
    timeLeft--;
    timer.innerText = timeLeft;
```

```
    if (timeLeft <= 0) { clearInterval(countdown);
```

```
        window.location.href = "/logout";
    }
    }, 1000);
</script>
</body>
</html>
```

index.html Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Smart Attendance</title>

<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap"
rel="stylesheet">

<style>

body{
margin:0;
height:100vh;
font-family:Poppins;
background:linear-gradient(135deg,#0f2027,#2c5364,#1c92d2);
background-size:400% 400%;
animation:bgMove 10s infinite alternate;
display:flex;
justify-content:center;
align-items:center;
color:white;
overflow:hidden;
}

/* 🔥 BACKGROUND ANIMATION */

@keyframes bgMove{
```

```
0%{background-position:0% 50%;}
100%{background-position:100% 50%;}
}
```

```
/*  TOP SCROLL TEXT */
```

```
.top-scroll{
position:absolute;
top:15px;
width:100%;
overflow:hidden;
white-space:nowrap;
font-size:26px;
font-weight:bold;
letter-spacing:3px;
color:#00e5ff;
}
```

```
.top-scroll span{
display:inline-block;
padding-left:100%;
animation:scroll 12s linear infinite;
}
```

```
/*  BOTTOM SCROLL TEXT */
```

```
.bottom-scroll{
position:absolute;
bottom:15px;
width:100%;
overflow:hidden;
white-space:nowrap;
font-size:20px;
font-weight:bold;
```

```
color:#ffd700;
}

.bottom-scroll span{
display:inline-block;
padding-left:100%;
animation:scroll 15s linear infinite;
}

/* SCROLL ANIMATION */
@keyframes scroll{
0%{transform:translateX(0);}
100%{transform:translateX(-100%);}
}

/* 💎 GLASS CARD */
.card{
background:rgba(255,255,255,0.1);
padding:50px;
border-radius:25px;
backdrop-filter:blur(15px);
text-align:center;
width:350px;
box-shadow:0 0 40px rgba(0,0,0,0.5);
animation:fadeIn 1.2s ease;
}

/* ✨ ENTRY ANIMATION */
@keyframes fadeIn{
from{opacity:0; transform:translateY(30px);}
to{opacity:1;}
}

.card h2{
```

```
margin-bottom:30px;
}

/* ○ BUTTONS */

button{
width:100%;
padding:15px;
margin:12px 0;
border:none;
border-radius:30px;
font-size:16px;
cursor:pointer;
transition:0.3s;
}

/* ● FACULTY BUTTON */

.login{
background:linear-gradient(45deg,#ff4b5c,#ff0066);
color:white;
box-shadow:0 0 20px rgba(255,0,102,0.6);
}

.login:hover{
transform:scale(1.08);
box-shadow:0 0 35px rgba(255,0,102,1);
}

/* ● STUDENT BUTTON */

.report{
background:linear-gradient(45deg,#00c9a7,#92fe9d);
color:black;
box-shadow:0 0 20px rgba(0,255,200,0.6);
}
```

```
.report:hover{
transform:scale(1.08);
box-shadow:0 0 35px rgba(0,255,200,1);
}
</style>
</head>
<body>
<!-- 📌 TOP TITLE -->
<div class="top-scroll">
<span>SMART ATTENDANCE SYSTEM • SMART ATTENDANCE SYSTEM • SMART
ATTENDANCE SYSTEM</span>
</div>
<!-- 💎 CARD -->
<div class="card">
<h2>Smart Attendance System</h2>
<a href="{{ url_for('login') }}">
<button class="login">Faculty Login</button>
</a>
<a href="{{ url_for('student_report') }}">
<button class="report">Student Report</button>
</a>
</div>
<!-- 🚩 FOOTER WATERMARK -->
<div class="bottom-scroll">
<span>DEVELOPED BY • J LAKSHMAN CHOWDARY • A LIKHITH • K SAKETH
CHOWDARY </span>
</div>
</body>
</html>
```

login.html Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Login</title>

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;600&display=swap"
rel="stylesheet">

<style>

body{
margin:0;
height:100vh;
font-family:Poppins;
display:flex;
}

/* 🟡 LEFT PANEL */

.left{
width:50%;
background:linear-gradient(135deg,#facc15,#f59e0b);
color:#1e293b;
display:flex;
flex-direction:column;
justify-content:center;
align-items:center;
text-align:center;
padding:40px;
}

/* LOGO */

.logo{
```

```
font-size:50px;
margin-bottom:10px;
animation:pop 0.8s ease;
}
```

```
@keyframes pop{
0%{transform:scale(0);}
100%{transform:scale(1);}
}
```

```
/* TITLE */
```

```
.left h1 {
font-size:30px;
margin-bottom:10px;
}
```

```
/* FEATURES */
```

```
.features{
margin-top:20px;
text-align:left;
font-size:16px;
line-height:2;
animation:fadeUp 1.2s ease;
}

.features p{
margin:6px 0;
font-weight:500;
opacity:0;
animation:slideIn 0.6s forwards;
}
```

```
/* STAGGER ANIMATION */

.features p:nth-child(1){animation-delay:0.2s;}
.features p:nth-child(2){animation-delay:0.4s;}
.features p:nth-child(3){animation-delay:0.6s;}
.features p:nth-child(4){animation-delay:0.8s;}
.features p:nth-child(5){animation-delay:1s;}
.features p:nth-child(6){animation-delay:1.2s;}

@keyframes slideIn{
from{opacity:0; transform:translateX(-20px);}
to{opacity:1; transform:translateX(0);}
}

@keyframes fadeUp{
from{opacity:0; transform:translateY(20px);}
to{opacity:1;}
}

/* RIGHT PANEL */

.right{
width:50%;
display:flex;
justify-content:center;
align-items:center;
background:#0f172a;
}

/* CARD */

.card{
background:#1e293b;
padding:40px;
```

```
border-radius:14px;
width:320px;
box-shadow:0 10px 30px rgba(0,0,0,0.4);
text-align:center;
color:white;
}
```

```
h2{
margin-bottom:25px;
}
```

```
/* INPUT */
```

```
.input-group{
position:relative;
margin-top:18px;
}
```

```
input{
width:100%;
padding:12px;
border-radius:8px;
border:none;
outline:none;
background:#334155;
color:white;
}
```

```
/* LABEL */
```

```
label{
position:absolute;
left:12px;
top:50%;
transform:translateY(-50%);
```

```
font-size:14px;
transition:0.3s;
color:#aaa;
}
input:focus + label,
input:not(:placeholder-shown) + label{
top:-8px;
font-size:12px;
color:#3b82f6;
}
/* BUTTON */
button{
width:100%;
padding:12px;
margin-top:25px;
border:none;
border-radius:8px;
background:#3b82f6;
color:white;
cursor:pointer;
transition:0.3s;
}
button:hover{
background:#2563eb;
}
/* MOBILE */
@media(max-width:768px){
.left{display:none;}
.right{width:100%;}
```

```
}
</style>

</head>

<body>

<!-- LEFT -->

<div class="left">

<div class="logo"> 🎓 </div>

<h1>Smart Attendance System</h1>

<div class="features">

<p>✓ Easy Attendance Marking</p>

<p>✓ Real-Time Student Reports</p>

<p>✓ Subject-wise Analytics</p>

<p>✓ Monthly Excel Reports</p>

<p>✓ Low Attendance Alerts</p>

<p>✓ Secure Login System</p>

</div>

</div>

<!-- RIGHT -->

<div class="right">

<div class="card">

<h2>Login</h2>

<form method="POST">

<div class="input-group">

<input type="text" name="code" required placeholder=" ">

<label>User ID</label>

</div>

<div class="input-group">

<input type="password" name="password" required placeholder=" ">

<label>Password</label>
```

```
</div>

<button>Login</button>

</form>

{% if error %}

<p style="color:red">{{error}}</p>

{% endif %}

</div>

</div>

</body>

</html>
```

dashboard.html Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Dashboard</title>


<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;500;700&display=swap"
rel="stylesheet">


<style>

body{

margin:0;

font-family:Poppins;

background:#0f172a;

color:white;

overflow:hidden;

}
```

```
/* ===== LOADER ===== */

#loader{
position:fixed;
width:100%;
height:100%;
background:#0f172a;
display:flex;
justify-content:center;
align-items:center;
z-index:9999;
}

.loader-circle{
width:60px;
height:60px;
border:6px solid #38bdf8;
border-top:6px solid transparent;
border-radius:50%;
animation:spin 1s linear infinite;
}

@keyframes spin{
100%{transform:rotate(360deg);}
}

/* ===== SIDEBAR ===== */

.sidebar{
width:230px;
height:100vh;
background:#020617;
position:fixed;
padding:20px;
```

```
}
```

```
.sidebar h3{  
color:#38bdf8;  
margin-bottom:30px;  
}
```

```
.sidebar a{  
display:block;  
color:white;  
padding:12px;  
margin:10px 0;  
border-radius:10px;  
text-decoration:none;  
transition:0.3s;  
}
```

```
.sidebar a:hover{  
background:#38bdf8;  
}
```

```
/* ===== MAIN ===== */
```

```
.main{  
margin-left:250px;  
padding:30px;  
}
```

```
/* ===== CARDS ===== */
```

```
.card-ui{  
background:rgba(255,255,255,0.05);  
padding:25px;
```

```

border-radius:20px;

text-align:center;

transition:0.3s;

animation:fadeIn 1s ease;

}


.card-ui:hover{

transform:translateY(-5px);

background:rgba(255,255,255,0.1);

}

@keyframes fadeIn{

from{opacity:0; transform:translateY(20px);}

to{opacity:1;}

}

/* ===== BUTTON ===== */

.btn-glow{

background:#38bdf8;

border:none;

color:black;

padding:10px;

border-radius:10px;

}

</style>

</head>

<body>

<!--  LOADER -->

<div id="loader">

    <div class="loader-circle"></div>

</div>

```

<!-- 🇮🇹 SIDEBAR -->

<div class="sidebar">

<h3> 🇮🇹 Dashboard</h3>

 🏠 Home

 📝 Attendance

 📊 Reports

<hr>

 🚪 Logout

</div>

<!-- 📦 MAIN CONTENT -->

<div class="main">

<h2>Welcome {{ faculty.name if faculty else "Admin" }}</h2>

<div class="row mt-4">

<div class="col-md-4">

<div class="card-ui">

<h4> 📝 Mark Attendance</h4>

Open

</div>

</div>

<div class="col-md-4">

<div class="card-ui">

<h4> 📊 Reports</h4>

View

</div>

</div>

<div class="col-md-4">

<div class="card-ui">

<h4> 📄 Excel Report</h4>

<select id="month" class="form-control mb-2">

```
<option value="1">January</option>
<option value="2">February</option>
<option value="3">March</option>
<option value="4">April</option>
<option value="5">May</option>
<option value="6">June</option>
<option value="7">July</option>
<option value="8">August</option>
<option value="9">September</option>
<option value="10">October</option>
<option value="11">November</option>
<option value="12">December</option>
</select>

<button onclick="download()" class="btn btn-glow w-100">Download</button>

</div>

</div>

</div>

</div>

<script>
// LOADER HIDE
window.onload = function(){
    document.getElementById("loader").style.display="none";
}

// DOWNLOAD
function download(){
    let m=document.getElementById("month").value;
    window.location="/export?month="+m;
}
</script>
```

</body>

</html>

Style.css Code:

```
/* ===== BODY ===== */
```

```
body {  
    margin: 0;  
    font-family: 'Segoe UI', sans-serif;  
    background: linear-gradient(135deg, #061a23, #0d2c3a);
```

```
/* 🔥 CENTER CONTENT */
```

```
display: flex;  
justify-content: center;  
align-items: center;  
height: 100vh;  
}
```

```
/* ===== SCROLLING WATERMARK COMMON ===== */
```

```
.scroll-bar {  
    position: fixed;  
    width: 100%;  
    overflow: hidden;  
    white-space: nowrap;  
    background: black;  
    color: yellow;  
    height: 35px;  
    line-height: 35px;  
    z-index: 9999;  
    font-weight: bold;  
    letter-spacing: 2px;  
}
```

```
/* Top position */
```

```
.top-bar {  
    top: 0;  
}
```

```
/* Bottom position */
```

```
.bottom-bar {  
    bottom: 0;  
}
```

```
/* Scrolling animation */
```

```
.scroll-bar span {  
    display: inline-block;  
    padding-left: 100%;  
    animation: marquee 15s linear infinite;  
}
```

```
@keyframes marquee {
```

```

    0% { transform: translateX(0); }
    100% { transform: translateX(-100%); }
}

/* ===== NAVBAR ===== */
.navbar {
    margin-top: 35px;
    background: #1f2e3a;
    padding: 15px;
    color: white;
    text-align: center;
    font-size: 20px;
    font-weight: bold;
}

/* ===== CONTAINER ===== */
.container {
    width: 350px;
    padding: 40px;
    background: white;
    border-radius: 15px;
    box-shadow: 0 15px 35px rgba(0,0,0,0.5);
    text-align: center;
}

/* ===== INPUTS ===== */
input, select {
    width: 90%;
    padding: 10px;
    margin: 10px 0;
    border-radius: 8px;
    border: 1px solid #ccc;
}

/* ===== BUTTONS ===== */
button {
    background: linear-gradient(45deg, #ff3c3c, #ff0066);
    color: white;
    border: none;
    padding: 10px 20px;
    margin: 10px;
    border-radius: 30px;
    cursor: pointer;
    transition: 0.3s ease;
}

button:hover {
    transform: scale(1.05);
    box-shadow: 0 0 15px rgba(255,0,100,0.7);
}

```

```

/* ===== TABLE ===== */
table {
    width: 100%;
    margin-top: 20px;
    border-collapse: collapse;
}

th {
    background: #1f2e3a;
    color: white;
    padding: 10px;
}

td {
    padding: 8px;
    border-bottom: 1px solid #ddd;
}

.low {
    color: red;
    font-weight: bold;
}

.good {
    color: green;
    font-weight: bold;
}

.error {
    color: red;
}

```

App.py Code:

```

from flask import Flask, render_template, request, redirect, url_for, session, send_file
import sqlite3
from datetime import date
import openpyxl
import smtplib
from email.mime.text import MIMEText
from apscheduler.schedulers.background import BackgroundScheduler
import os

app = Flask(__name__)
app.secret_key = "chowdary8222"

# ===== DB =====
def get_db():
    conn = sqlite3.connect("attendance.db")
    conn.row_factory = sqlite3.Row
    return conn

```

```
# ===== CREATE TABLES =====
```

```
def create_tables():
```

```
    db = get_db()
```

```
    cursor = db.cursor()
```

```
    cursor.execute("""
```

```
    CREATE TABLE IF NOT EXISTS students (
```

```
        student_id TEXT PRIMARY KEY,
```

```
        name TEXT,
```

```
        email TEXT
```

```
    )
```

```
    """)
```

```
    cursor.execute("""
```

```
    CREATE TABLE IF NOT EXISTS faculty (
```

```
        faculty_code TEXT PRIMARY KEY,
```

```
        name TEXT,
```

```
        password TEXT,
```

```
        subject TEXT
```

```
    )
```

```
    """)
```

```
    cursor.execute("""
```

```
    CREATE TABLE IF NOT EXISTS attendance (
```

```
        id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
        student_id TEXT,
```

```
        faculty_code TEXT,
```

```
        subject TEXT,
```

```
        date TEXT,
```

```
        hour TEXT,
```

```
        status TEXT
```

```
    )
```

```
    """)
```

```
    cursor.execute("""
```

```
    CREATE TABLE IF NOT EXISTS email_logs (
```

```
        student_id TEXT,
```

```
        last_sent TEXT
```

```
    )
```

```
    """)
```

```
    db.commit()
```

```
create_tables()
```

```
# ===== EMAIL =====
```

```
def send_email(to_email, name, percentage):
```

```
    sender = "lakshmanchowdary2005@gmail.com"
```

```
    password = "vhmwwqjwsqdrwlhu"
```

```
msg = MIMEText(f"""\nDear {name},
```

```


Your attendance is {percentage}%.
```

```


This is below 75%.
```

```


Please attend classes regularly.
```

```


Regards,\nHOD of IT Department\n""")
```

```
msg['Subject'] = "Low Attendance Alert"\nmsg['From'] = sender\nmsg['To'] = to_email
```

```
try:\n    server = smtplib.SMTP("smtp.gmail.com", 587)\n    server.starttls()\n    server.login(sender, password)\n    server.send_message(msg)\n    server.quit()\nexcept Exception as e:\n    print("Email Error:", e)
```

```
# ===== EMAIL ONCE =====
```

```
def send_email_once(student_id, email, name, percentage):\n    db = get_db()\n    cursor = db.cursor()\n\n    today = str(date.today())\n\n    cursor.execute("SELECT * FROM email_logs WHERE student_id=?", (student_id,))\n    record = cursor.fetchone()\n\n    if record and record["last_sent"] == today:\n        return\n\n    send_email(email, name, percentage)\n\n    if record:\n        cursor.execute("UPDATE email_logs SET last_sent=? WHERE student_id=?",\n                        (today, student_id))\n    else:\n        cursor.execute("INSERT INTO email_logs VALUES (?,?)",\n                        (student_id, today))\n\n    db.commit()
```

```

# ===== DAILY EMAIL =====
def send_daily_alerts():
    db = get_db()
    cursor = db.cursor()

    cursor.execute("SELECT * FROM students")
    students = cursor.fetchall()

    for student in students:
        sid = student["student_id"]

        cursor.execute("SELECT COUNT(*) as total FROM attendance WHERE student_id=?",
(sid,))
        total = cursor.fetchone()["total"]

        cursor.execute("SELECT COUNT(*) as present FROM attendance WHERE student_id=?
AND status='Present'", (sid,))
        present = cursor.fetchone()["present"]

        percentage = (present / total * 100) if total else 0

        if percentage < 75 and student["email"]:
            send_email_once(sid, student["email"], student["name"], round(percentage, 2))

# ===== SCHEDULER =====
scheduler = BackgroundScheduler()
scheduler.add_job(send_daily_alerts, 'cron', hour=9, minute=0)
scheduler.start()

# ===== HOME =====
@app.route("/")
def home():
    return render_template("index.html")

# ===== LOGIN =====
@app.route("/login", methods=["GET", "POST"])
def login():
    if request.method == "POST":
        code = request.form["code"]
        password = request.form["password"]

        db = get_db()
        cursor = db.cursor()

        if code == "admin" and password == "admin123":
            session["role"] = "admin"
            return redirect("/dashboard")

        cursor.execute("SELECT * FROM faculty WHERE faculty_code=? AND password=?",
(code, password))

```

```

faculty = cursor.fetchone()

    if faculty:
        session["role"] = "faculty"
        session["faculty_code"] = code
        return redirect("/dashboard")

    return render_template("login.html", error="Invalid Login")

return render_template("login.html")

# ===== DASHBOARD =====
@app.route("/dashboard")
def dashboard():
    if "role" not in session:
        return redirect("/login")

    db = get_db()
    cursor = db.cursor()

    faculty = None
    if session["role"] == "faculty":
        cursor.execute("SELECT * FROM faculty WHERE faculty_code=?",
(session["faculty_code"],))
        faculty = cursor.fetchone()

    return render_template("dashboard.html", faculty=faculty)

# ===== MARK ATTENDANCE =====
@app.route("/mark_attendance", methods=["GET", "POST"])
def mark_attendance():
    if session.get("role") != "faculty":
        return redirect("/login")

    db = get_db()
    cursor = db.cursor()

    cursor.execute("SELECT * FROM faculty WHERE faculty_code=?",
(session["faculty_code"],))
    faculty = cursor.fetchone()

    cursor.execute("SELECT student_id,name FROM students ORDER BY student_id")
    students = cursor.fetchall()

    if request.method == "POST":
        hour = request.form["hour"]
        absent = request.form.getlist("absent")

        for s in students:
            sid = s["student_id"]

```

```

status = "Absent" if sid in absent else "Present"

        cursor.execute(
            "SELECT 1 FROM attendance WHERE student_id=? AND subject=? AND date=?
AND hour=?",
            (sid, faculty["subject"], str(date.today()), hour)
        )
        exists = cursor.fetchone()

        if not exists:
            cursor.execute("""
                INSERT INTO attendance
                (student_id, faculty_code, subject, date, hour, status)
                VALUES (?, ?, ?, ?, ?, ?)
            """, (sid, session["faculty_code"], faculty["subject"], str(date.today()), hour, status))

        db.commit()
        return render_template("success.html")

    return render_template("mark_attendance.html", students=students)

# ===== STUDENT REPORT =====
@app.route("/student_report", methods=["GET", "POST"])
def student_report():
    db = get_db()
    cursor = db.cursor()

    if request.method == "POST":
        sid = request.form["student_id"].upper()

        cursor.execute("SELECT * FROM students WHERE student_id=?", (sid,))
        student = cursor.fetchone()

        if not student:
            return render_template("student_report.html", error="Student not found", subjects=[],
present_counts=[])

        cursor.execute("SELECT COUNT(*) as total FROM attendance WHERE student_id=?",
(sid,))
        total = cursor.fetchone()["total"]

        cursor.execute("SELECT COUNT(*) as present FROM attendance WHERE student_id=?
AND status='Present'", (sid,))
        present = cursor.fetchone()["present"]

        cursor.execute("""
            SELECT subject,
            SUM(CASE WHEN status='Present' THEN 1 ELSE 0 END) as present
            FROM attendance
            WHERE student_id=?

```

```

GROUP BY subject
    """', (sid,))
    data = cursor.fetchall()

    subjects = []
    present_counts = []

    for d in data:
        subjects.append(d["subject"])
        present_counts.append(d["present"] if d["present"] else 0)

    percentage = (present / total * 100) if total else 0

    if percentage < 75 and student["email"]:
        send_email_once(sid, student["email"], student["name"], round(percentage, 2))

    return render_template("student_report.html",
                           student=student,
                           total=total,
                           present=present,
                           percentage=round(percentage, 2),
                           subjects=subjects,
                           present_counts=present_counts)

    return render_template("student_report.html", subjects=[], present_counts=[])

# ===== EXPORT =====
@app.route("/export")
def export():
    db = get_db()
    cursor = db.cursor()

    month = request.args.get("month")

    if not month:
        month = str(date.today().month)

    cursor.execute("""
        SELECT DISTINCT subject
        FROM attendance
        WHERE strftime('%m', date)=?
    """, (month.zfill(2),))
    subjects = [row["subject"] for row in cursor.fetchall()]

    cursor.execute("SELECT student_id, name FROM students ORDER BY student_id")
    students = cursor.fetchall()

    wb = openpyxl.Workbook()
    sheet = wb.active

```

```

header = ["Hallticket No"] + subjects + ["Total", "Percentage"]
sheet.append(header)

for student in students:
    sid = student["student_id"]
    row = [sid]

    total_all = 0
    present_all = 0

    for subject in subjects:
        cursor.execute("""
            SELECT COUNT(*) as total,
            SUM(CASE WHEN status='Present' THEN 1 ELSE 0 END) as present
            FROM attendance
            WHERE student_id=? AND subject=? AND strftime('%m', date)=?
            """, (sid, subject, month.zfill(2)))

        res = cursor.fetchone()
        total = res["total"] or 0
        present = res["present"] or 0

        row.append(f"{present}/{total}")
        total_all += total
        present_all += present

    row.append(f"{present_all}/{total_all}")
    percentage = (present_all / total_all * 100) if total_all else 0
    row.append(round(percentage, 2))

sheet.append(row)

#  COLOR PART
from openpyxl.styles import PatternFill

red = PatternFill(start_color="FF9999", end_color="FF9999", fill_type="solid")
green = PatternFill(start_color="99FF99", end_color="99FF99", fill_type="solid")

percent_col = len(subjects) + 3
cell = sheet.cell(row=sheet.max_row, column=percent_col)

if percentage < 75:
    cell.fill = red
else:
    cell.fill = green
file = "attendance.xlsx"
wb.save(file)
return send_file(file, as_attachment=True)

```

```
# ===== LOGOUT =====
```

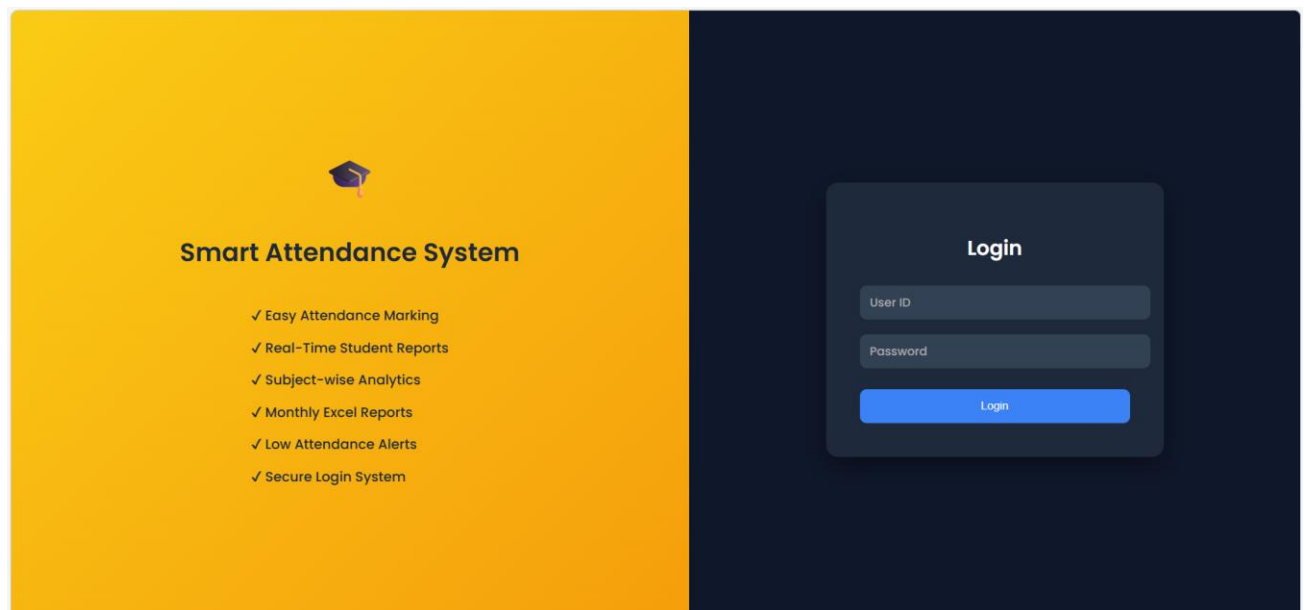
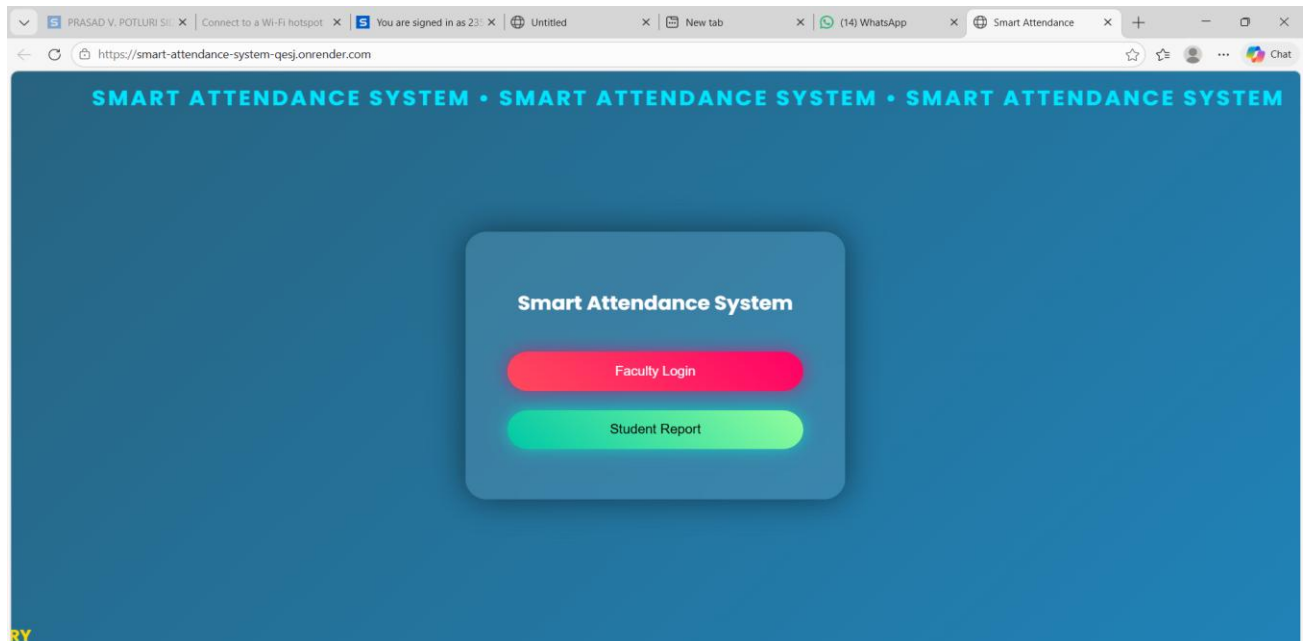
```

@app.route("/logout")
def logout():
    session.clear()
    return redirect("/")

# ===== RUN =====
if __name__ == "__main__":
    port = int(os.environ.get("PORT", 5000))
    app.run(host="0.0.0.0", port=port)

```

Outputs:



**Dashboard**

Home

Attendance

Reports

Logout

Welcome Mr. CH. PRANEETH

Mark Attendance

Open

Reports

View

Excel Report

January

Download

Mark Attendance

Back

Logout

Search student...

Present: 61

Absent: 11

Select All

Clear All

Select Hour:

Hour 3

Select Absent Students

23501A1201

ADARI KUSUMA

23501A1202

AKULA RUSHITHA

23501A1203

ALURI SREYA

23501A1204

ANDE LIKHITH

23501A1205

ANGAJALA DIVYA

23501A1206

APPIKATLA KEERTHANA

23501A1207

ARELLA GIRIDHAR

23501A1208

AVUTU MEGHANA

23501A1209

BAGGU MANASA

23501A1210

BANAVATH ANU BAI

23501A1211

BANDARU VOLINI PRIYA

23501A1212

BANDI DHANUSRI

23501A1213

BANDLAMUDI LAHARI

23501A1214

BANDUCHODAY NITESH KUMAR

23501A1215

BASAVA ROHITH REDDY

23501A1216

BEVARA LAKSHMI

23501A1217

BOTTA INKITA

23501A1218

CHAGANTIPATI NISHANTH SRIKAR

23501A1219

CHAKRAVADHANULA SAI SUSHMITHA

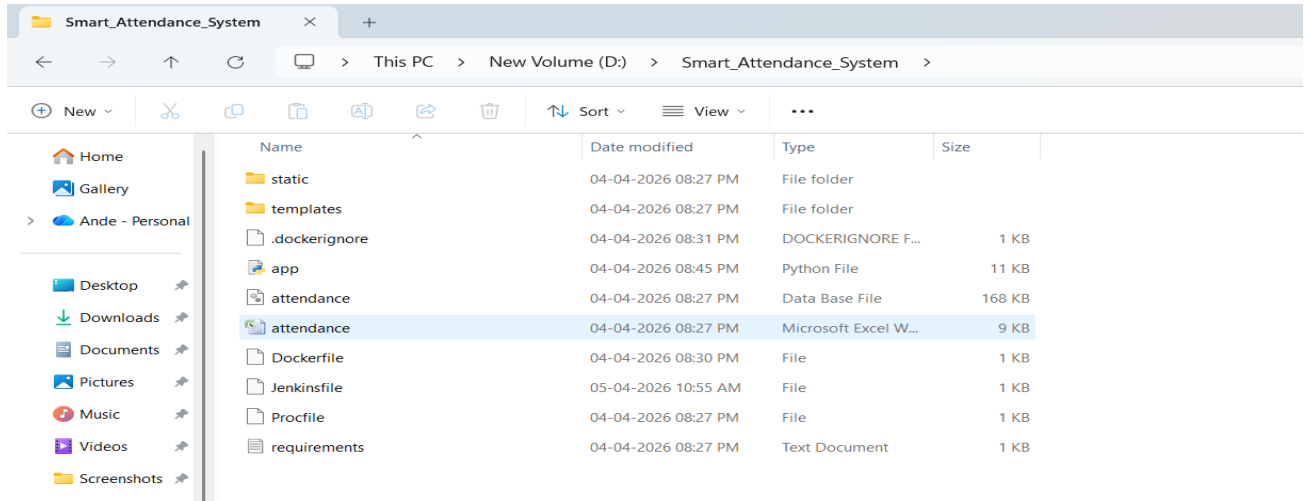
23501A1220

CHATRATHI SHANMUKHA PRASHANTH

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Hallticket No	DevOps	Software Project M.	Soft Skills	CRT	Machine Learning Lab	E-Waste Managemen	Cloud Computing	Cloud Computing Lab	Cryptography & Network S	Machine Learning	Total	Percentage	
2	23501A1201	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
3	23501A1202	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
4	23501A1203	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
5	23501A1204	2/3	3/4	2/3	3/4	2/2	2/2	1/3	1/2	1/4	1/1	18/28	64.29	
6	23501A1205	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
7	23501A1206	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
8	23501A1207	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
9	23501A1208	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
10	23501A1209	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
11	23501A1210	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
12	23501A1211	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
13	23501A1212	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
14	23501A1213	3/3	3/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	26/28	92.86	
15	23501A1214	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
16	23501A1215	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
17	23501A1216	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
18	23501A1217	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
19	23501A1218	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	26/28	92.86	
20	23501A1219	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
21	23501A1220	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
22	23501A1221	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
23	23501A1222	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
24	23501A1223	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
25	23501A1224	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
26	23501A1225	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
27	23501A1226	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
28	23501A1227	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
29	23501A1228	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
30	23501A1229	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
31	23501A1230	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
32	23501A1231	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	
33	23501A1232	3/3	4/4	3/3	3/4	2/2	2/2	3/3	2/2	4/4	1/1	27/28	96.43	

Dockerfile Configuration

Your Dockerfile likely contains instructions similar to this



Docker File:-

Use an official Python runtime as a parent image

FROM python:3.9-slim

Set the working directory in the container

WORKDIR /app

Copy the requirements file into the container

COPY requirements.txt .

Install any needed packages specified in requirements.txt

RUN pip install --no-cache-dir -r requirements.txt

Copy the rest of the application code

COPY . .

Expose the port Flask runs on (usually 5000)

EXPOSE 5000

Define environment variable for Flask

ENV FLASK_APP=app.py

Run the application

CMD ["python", "app.py"]

```
C:\Windows\System32\cmd.exe x + v
D:\Smart_Attendance_System>dir
Volume in drive D is New Volume
Volume Serial Number is 56CA-EBE1

Directory of D:\Smart_Attendance_System

04-04-2026 20:31 <DIR>          .
04-04-2026 20:31             42  .dockerignore
04-04-2026 20:27          11,201 app.py
04-04-2026 20:27        172,032 attendance.db
04-04-2026 20:27         8,518 attendance.xlsx
04-04-2026 20:30         557 Dockerfile
04-04-2026 20:27         21 Procfile
04-04-2026 20:27         46 requirements.txt
04-04-2026 20:27 <DIR>          static
04-04-2026 20:27 <DIR>          templates
04-04-2026 20:27 7 File(s)        192,417 bytes
                  3 Dir(s)    105,278,976,000 bytes free

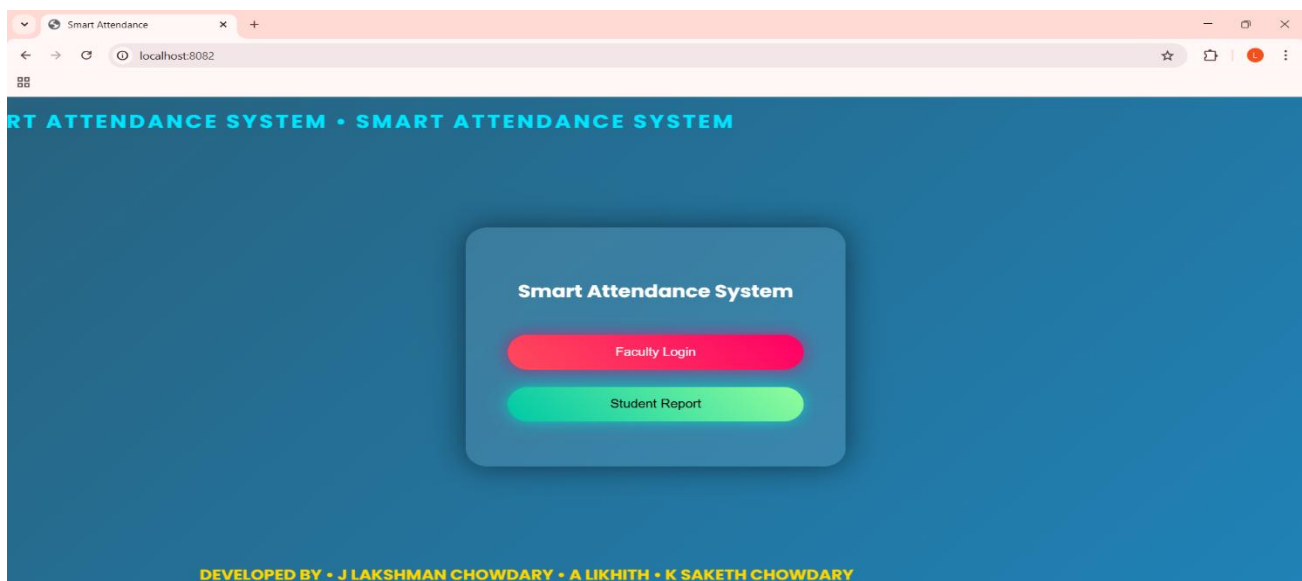
D:\Smart_Attendance_System>docker build -t smart-attendance-app .
[*] Building 1.8s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 596B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 82B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aablacdale13cf
=> [internal] load build context
=> => transferring context: 2.67kB
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:5a8a8272358e13a5e75113a587f69c1309235b128510689e7d4b5169664582ab
=> => exporting config sha256:ccf1dc10f8ffb7ab78777de627863094306823a10859b5427f2107e007a20a77
=> => exporting attestation manifest sha256:d7b26e75e96739f6d0666fa0c4a3f61131bb058825966828b013ae730581eaeab
docker: desktop-linux
```

Building and Running Locally

1. **Build the Image:** Open your terminal in the SMART-ATTENDANCE-SYSTEM directory and run: `docker build -t smart-attendance-system-app .`
2. **Run the Container:** `docker run -d -p 8082:5000 smart-attendance-system-app`

```
PS D:\Smart_Attendance_System> docker run -d -p 8082:5000 smart-attendance-app
5570c46b9177e0ccbc0823cff3e1cd3bc28293a5b10b039419d9d35e3e5cb215
PS D:\Smart_Attendance_System> S
```

3. **Access:** Open your browser and go to <http://localhost:8080>

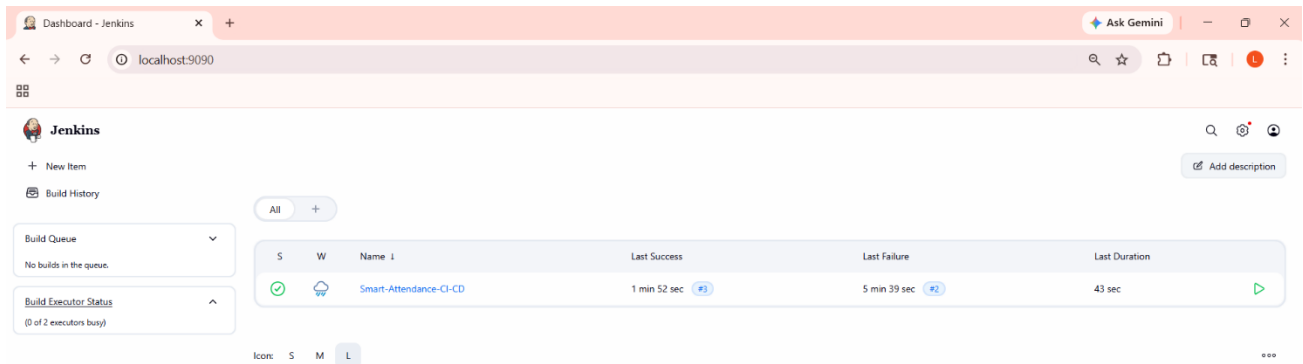


Jenkins CI/CD Pipeline

The `Jenkinsfile` in your directory automates the deployment process.

Detailed Pipeline Steps:

1. **Checkout Stage:** Jenkins pulls the latest code from your version control system (e.g., GitHub or a local directory).
2. **Build Stage:** Jenkins triggers the Docker build command to create a new image of the application.
 - o *Command:* `docker build -t smart-attendance-system:${env.BUILD_ID} .`



1. **Deployment Stage:** Stops any currently running container of the old version.
 - o Starts a new container using the freshly built image.
 - o *Command:* `docker run -d --name attendance-app-container -p 82:5000 smart-attendance-system:${env.BUILD_ID}`

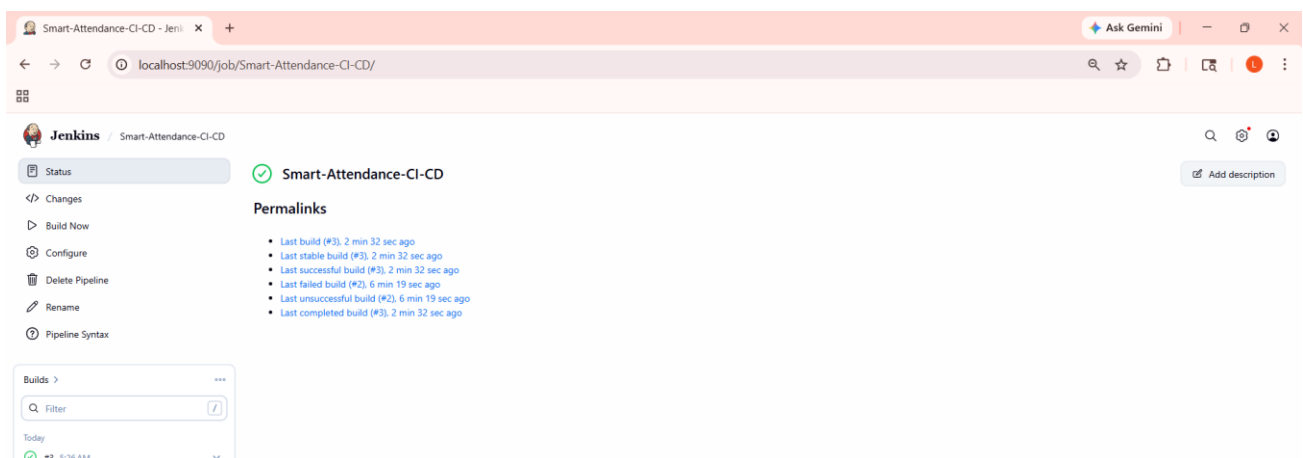
How to Trigger the Build:

1. Open your **Jenkins Dashboard**.
2. Create a **New Item** -> **Pipeline**.
3. Under the **Pipeline** section, select "Pipeline script from SCM."
4. Point it to your project directory. Jenkins will automatically read your `Jenkinsfile` and execute the steps.

Directory Structure

Smart_Attendance_System/

```
|— app/                # All Python/Flask logic goes here
| |— static/          # CSS, JS, and Images
| | |— css/
| | |— js/
| |— templates/       # HTML files (login.html, dashboard.html)
| |— __init__.py
| |— routes.py         # Flask routes and logic
| |— models.py         # Database models (if using SQLAlchemy)
|— tests/             # Unit tests for your code
| |— test_app.py
|— instance/          # Local database files (e.g., SQLite)
|— .gitignore          # Files to ignore in Git (venv, __pycache__)
|— Dockerfile          # Instructions to build the Docker image
|— Jenkinsfile         # Instructions for the Jenkins Pipeline
|— requirements.txt    # List of Python dependencies (Flask, etc.)
|— run.py              # The entry point to start your application
```



Conclusion :

The Smart Attendance System successfully demonstrates how modern web technologies can be used to automate and improve traditional attendance management processes. The application provides an efficient, accurate, and user-friendly platform for recording, tracking, and analyzing student attendance. By eliminating manual errors and reducing administrative workload, the system enhances productivity and ensures reliable data management.

The integration of features such as real-time attendance marking, automated percentage calculation, Excel report generation with conditional formatting, and email alerts for low attendance adds significant value to the system. The use of Flask and SQLite makes the application lightweight, easy to deploy, and suitable for academic environments.

Overall, the project achieves its objective of creating a scalable and practical solution for attendance management. It also highlights the importance of combining frontend, backend, and database technologies to build a complete and functional web application. This system can be further enhanced with additional features such as biometric integration, mobile app support, and advanced analytics, making it adaptable for future improvements.

The **Smart Attendance System** web application successfully demonstrates a modern approach to web development and deployment. By moving beyond a simple static site and integrating **Docker** and **Jenkins**, the project achieves a high level of professional reliability.

GitHub Repository:-

https://github.com/likhithande/Smart_Attendance_System

Deployment Link:-

<https://smart-attendance-system-qesj.onrender.com/>

The above link opens the deployed Smart Attendance System .The source code is available in the listed GitHub repositories.